MedInsight: Evaluating Retrieval-Augmented Vision-Language Models for Evidence-Grounded Medical Image Understanding

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***Abstract*—This paper investigates whether retrieval-augmented prompting improves vision-language model (VLM) accuracy on medical visual question answering (VQA), using only openly licensed, non-credentialed data. We build a CLIP-embedded FAISS retrieval index over the ROCOv2 radiology corpus (approximately 90,000 image-caption pairs) and use it to supply retrieved evidence captions to a pretrained BLIP-2 vision-language model, comparing this retrieval-augmented condition against the same model answering with no retrieved evidence. Both conditions share identical tokenization, decoding parameters, and scoring code, differing only in prompt content, and are evaluated on the official VQA-RAD held-out test split (451 examples: 251 closed-ended, 200 open-ended) using BLEU-4, ROUGE-L, and exact match, together with paired bootstrap significance testing. Retrieval augmentation improved every reported metric at the default retrieval depth ($k = 5$), with closed-question accuracy rising from 43.0% to 47.8% and overall exact match from 26.8% to 29.5%; however, neither improvement was statistically significant (closed-question exact match: $p = 0.1848$, 95% CI $[-0.020, 0.116]$; open-question ROUGE-L: $p = 0.9642$, 95% CI $[-0.032, 0.033]$; $n = 451$). A secondary finding from a sweep over $k \in \{1, 3, 5, 10\}$ showed accuracy peaking at $k = 1$ and declining monotonically as retrieval depth increased, suggesting that excess retrieved evidence can dilute rather than reinforce a base model's answer. These results indicate retrieval augmentation shows a promising but unconfirmed effect on medical VQA at this scale, with retrieval depth mattering at least as much as the presence of retrieval itself.**

***Index Terms*—retrieval-augmented generation, vision-language models, medical visual question answering, radiology, evaluation**

I. Introduction

Vision-language models (VLMs) that answer free-text questions about images have progressed rapidly on general-domain benchmarks, driven by large-scale image-text pretraining [1] and efficient adaptation of frozen encoders and language models [2]. Applying these models to medical imaging is attractive: a system that can answer a clinician's question about a chest radiograph or CT slice could support triage, education, and second-opinion workflows. However, general-domain VLMs are trained overwhelmingly on natural images and web-scale captions, and even biomedical-specific variants [3] are trained on a fixed snapshot of literature and cannot cite the specific evidence behind a given answer. An answer generated purely from a model's parameters is difficult to audit: a clinician evaluating the system's output has no way to inspect what, if anything, the model's answer was based on.

Retrieval-augmented generation (RAG) offers one route to addressing this. Originally proposed for knowledge-intensive text tasks [4], RAG couples a parametric generator with a non-parametric memory: instead of answering from weights alone, the model is given retrieved supporting evidence and conditions its answer on that evidence. In principle this offers two benefits for medical VQA specifically: (1) the retrieved evidence is inspectable, giving a clinician something concrete to check the answer against, and (2) the model may produce more accurate answers by grounding its response in similar, verified cases rather than relying solely on what it memorized during pretraining.

Whether retrieval augmentation actually improves accuracy on medical VQA, however, is an empirical question, not a foregone conclusion. Retrieved evidence could just as easily distract a model, dilute its attention across irrelevant content, or be ignored outright if the model was never trained to make use of retrieved context. Answering this question rigorously requires a controlled comparison: the same base model, the same test questions, the same generation parameters, differing only in whether retrieved evidence is present in the prompt.

A. Contributions

This paper makes three contributions:

1) An open, reproducible pipeline for retrieval-augmented medical VQA built entirely on openly licensed, non-credentialed datasets — ROCOv2 [5] as the retrieval corpus and VQA-RAD [6] as the evaluation benchmark — so the full pipeline can be reproduced without a data use agreement.
2) A controlled empirical comparison of a vision-language model with and without retrieved evidence, using the same checkpoint, generation parameters, and scoring code for both conditions (Section IV), evaluated with

both standard automatic metrics and paired statistical significance testing (Section V).
3) A documented, config-driven, tested codebase intended to be extensible to larger or credentialed corpora (e.g., MIMIC-CXR) in future work, released alongside this paper.

II. Related Work

A. Vision-Language Models

Contrastive image-text pretraining, exemplified by CLIP [1], learns a shared embedding space for images and text from large-scale web data without task-specific labels, and underlies the image encoder used in this work's retriever (Section IV). Building on such encoders, generative vision-language models couple a frozen or lightly-adapted vision encoder with a large language model to produce free-text answers conditioned on an image. BLIP-2 [2] bridges a frozen image encoder and a frozen language model with a lightweight Querying Transformer, substantially reducing the number of trainable parameters relative to end-to-end pretraining. LLaVA [7] instead couples a vision encoder to a language model through visual instruction tuning on model-generated instruction-following data. This project's baseline VLM (Section IV) is drawn from this family of pretrained, general-domain checkpoints.

B. Medical Vision-Language Models

General-domain VLMs transfer imperfectly to medical imaging, where visual patterns, terminology, and the stakes of an incorrect answer all differ from natural-image domains. LLaVA-Med [3] adapts the LLaVA recipe to biomedicine, continuing training on figure-caption pairs drawn from PubMed Central and biomedical instruction-following data generated with the assistance of a large language model, and reports improvements over general-domain VLMs on medical visual question answering. This line of work establishes that domain adaptation helps, but the resulting models still answer purely from their trained parameters at inference time — they do not retrieve or cite supporting evidence for a specific query, which is the gap this paper's retrieval-augmented condition addresses.

C. Retrieval-Augmented Generation

Retrieval-augmented generation was introduced for knowledge-intensive text tasks [4], combining a pretrained sequence-to-sequence generator with a dense vector index over a non-parametric corpus, retrieved via maximum inner-product search. The approach's central premise — grounding generation in retrieved, inspectable evidence rather than relying solely on parametric memory — transfers naturally to a setting where evidence is diagnostically meaningful and open to human review, as in medical imaging. Efficient large-scale nearest-neighbor search over dense embeddings, which the retrieval side of this pipeline depends on, is provided by FAISS [8], used here as an in-process CPU-based flat inner-product index (Section IV) appropriate for a corpus of ROCOv2's scale.

TABLE I
DATASETS USED IN THIS WORK.

Role	Dataset	License
Retrieval corpus	ROCOv2 [5]	CC BY-NC-SA 4.0
Evaluation benchmark	VQA-RAD [6]	CC0 (public domain)

D. Medical VQA Benchmarks

VQA-RAD [6] is the first manually constructed medical VQA dataset in which clinicians posed naturally occurring questions about radiology images and supplied reference answers, spanning both closed-ended (yes/no-style) and open-ended free-text questions across CT, MRI, and X-ray studies. Its clinician-authored, non-synthetic questions and small, well-defined test split make it a natural fit for the retrieval-augmented comparison in this paper, at the cost of a test-split size small enough to require explicit statistical significance testing (Section V) before treating any accuracy delta as reliable.

E. Positioning of This Work

Prior work has separately established that (a) domain-adapted VLMs outperform general-domain VLMs on medical VQA [3], and (b) retrieval augmentation improves knowledge-intensive text generation in non-medical settings [4]. This paper combines these two threads directly: rather than adapting a model's weights to the medical domain, it tests whether providing retrieved medical evidence *at inference time*, to an otherwise unmodified VLM, produces a measurable accuracy improvement on medical VQA — and, if so, where that improvement does and does not appear (Section VII).

III. Datasets

This work uses two openly licensed, non-credentialed datasets, each serving a distinct role in the pipeline: one as the corpus the retriever searches over, the other as the held-out benchmark the downstream task is graded on. Table I summarizes both; full preprocessing detail is in the project's `docs/dataset.md`.

A. ROCOv2: Retrieval Corpus

ROCOv2 [5] pairs approximately 90,000 radiological images with captions extracted from the PMC Open Access subset, spanning multiple modalities (CT, MRI, X-ray, and ultrasound). Each caption is a clinically written figure caption from published biomedical literature, not a crowd-sourced description. This corpus plays the role of the non-parametric evidence store in the retrieval-augmented condition (Section IV): at inference time, the query image is embedded and used to retrieve the most visually similar corpus entries, whose captions are then supplied to the model as evidence.

One caveat carried through to the retrieval design: ROCOv2 captions are figure captions, not full clinical reports, and occasionally reference comparative or positional context (e.g., a prior figure panel) that is not recoverable from the isolated retrieved image. Section VII returns to whether this measurably affects retrieved-evidence quality.

B. VQA-RAD: Evaluation Benchmark

VQA-RAD [6] contains 3,515 question-answer pairs over 315 radiology images, with questions written directly by clinicians rather than crowdworkers, covering both closed-ended (yes/no-style) and open-ended free-text questions across CT, MRI, and X-ray studies. Using a benchmark disjoint from the retrieval corpus is deliberate: it evaluates the downstream VQA task independently of the retriever's own source corpus, rather than measuring the model's ability to recover an answer key it was given evidence from.

C. Splitting and Preprocessing

VQA-RAD is evaluated on its own official held-out test split (451 examples), rather than a further internal re-split of the training portion, so that results are directly comparable to other published VQA-RAD baselines that use the same standard split. The official training portion (1,793 examples) is further divided 90/10 into train/validation internally, stratified by answer type (closed vs. open) where that label is available. The Hugging Face mirror of VQA-RAD used here (`flaviagiammarino/vqa-rad`) does not expose an explicit closed/open answer-type field; this label is instead recovered from the answer text itself, treating a normalized answer of exactly "yes" or "no" as closed-ended and every other answer as open-ended. This is an approximation of the original VQA-RAD annotation – a small number of closed-ended, non-yes/no questions (e.g. "which side is the lesion on?") will be mislabeled open-ended – and is noted as a limitation in Section VIII. ROCOv2 is not split, since it serves only as the full retrieval corpus, not a held-out evaluation set. Images are resized to 224×224 and normalized with standard ImageNet statistics, noted here as a documented default rather than a finding specific to medical images, which have different intensity distributions than natural images; captions are truncated to 128 whitespace-delimited tokens before being written to the retrieval manifest.

D. Why Not MIMIC-CXR

MIMIC-CXR is the chest radiograph dataset most commonly used in published medical VLM work. It was not used here because it requires a credentialed PhysioNet access process (CITI training and institutional affiliation review) with a latency of one to several weeks, which would have delayed every downstream milestone of a timeline-constrained project. ROCOv2 and VQA-RAD require no credentialing and no protected health information handling, allowing the full pipeline to be built and validated end to end first. Extending the retrieval corpus to include MIMIC-CXR, once credentialed access is obtained, is identified as future work (Section VIII).

IV. Method

The pipeline consists of three components, built and tested independently: a baseline vision-language model with no retrieved evidence, an image retriever over the ROCOv2 corpus, and a retrieval-augmented model that composes the first two. All three share code where possible, which is the design choice that makes the baseline-vs-RAG comparison in Section VI valid rather than an apples-to-oranges benchmark. Figure 1 summarizes both conditions end to end.

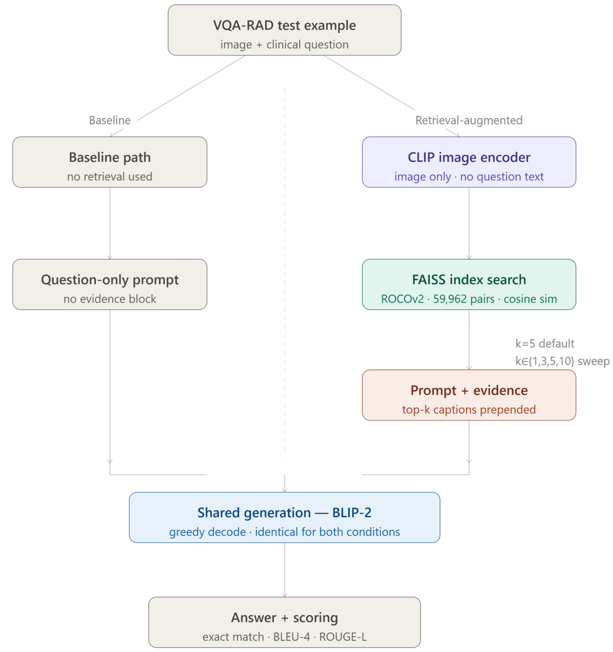

Fig. 1. MedInsight pipeline. Both conditions share identical tokenization, decoding, and scoring code (`BaselineVLM.generate_from_prompt`); the retrieval-augmented path differs only in that retrieved ROCOv2 captions are prepended to the prompt before generation.

A. Baseline Vision-Language Model

The baseline condition uses a pretrained, general-domain vision-language model checkpoint (a BLIP-2-family model [2], configured in `configs/model_config.yaml`) with no modification and no retrieved context: it is given an image and a question, and generates a free-text answer directly. This is deliberately the simplest possible condition — it isolates the effect of retrieval augmentation from any effect of domain-specific fine-tuning, which prior work has already studied separately [3].

B. Retriever

The retriever embeds every ROCOv2 image with a CLIP image encoder [1], L2-normalizes the resulting vectors, and indexes them with a FAISS [8] flat inner-product index — exact (non-approximate) nearest-neighbor search, appropriate at ROCOv2's corpus scale (∼90K vectors) without requiring an approximate index such as IVF or HNSW. At query time, the same CLIP encoder embeds the query image, and the index returns the k most visually similar corpus entries by cosine similarity (equivalent to inner product over normalized vectors), together with their captions and similarity scores.

C. Retrieval-Augmented Model

The retrieval-augmented condition composes the baseline model and the retriever rather than reimplementing generation: it retrieves the top-k similar corpus captions for the query image, constructs a prompt that prepends those captions (truncated to a fixed word budget) before the original question, and passes that prompt through the *same* underlying generation call the baseline condition uses. Concretely, both conditions call one shared code path (`BaselineVLM.generate_from_prompt` in the released code) with the same tokenization, decoding parameters (deterministic decoding; see Section V), and post-processing — the only difference between the two experimental conditions is the text of the prompt itself. If no evidence is retrieved for a given query (e.g. a cold or empty index), the system falls back to the bare question with no evidence block, i.e. to baseline behavior, rather than emitting a malformed prompt.

D. Design Rationale: Shared Generation Code

This shared-code-path design is the paper's central methodological safeguard. A comparison between two differently-implemented pipelines — even ones intended to differ only in "using retrieval or not" — risks confounding the retrieval effect with incidental implementation differences (e.g. different tokenization, different truncation behavior, different decoding temperature). By construction, that confound is eliminated here: the two conditions differ in exactly one respect, the prompt text, which is what licenses attributing any measured accuracy difference (Section VI) to retrieval augmentation itself.

V. Experimental Setup

A. Independent Variable

The single independent variable across the two experimental conditions is the presence or absence of retrieved evidence in the prompt (baseline vs. retrieval-augmented; Section IV). All other factors — model checkpoint, decoding parameters, test examples, and scoring code — are held fixed between conditions by construction.

B. Model and Decoding Configuration

The baseline checkpoint is `Salesforce/blip2-opt-2.7b`, with the vision encoder frozen and the language model unfrozen (`configs/model_config.yaml`); `microsoft/llava-med-v1.5-mistral-7b` is retained as an alternate candidate to substitute if the primary checkpoint underperforms during initial validation. Generation is deterministic (greedy decoding, `do_sample=false`) with a maximum of 128 new tokens, so that repeated runs of either condition on the same input are reproducible. The retriever uses `openai/clip-vit-base-patch32` as its image encoder and returns the top k=5 most similar corpus items per query by default; Section VI additionally reports a sweep over $k \in \{1, 3, 5, 10\}$ to test the sensitivity of accuracy to this choice.

C. Evaluation Metrics

VQA-RAD questions are scored differently by type, following the distinction the dataset itself makes between closed-ended and open-ended questions [6]: closed-ended (yes/no-style) questions are scored by exact match after normalization (lowercasing, article and punctuation removal); open-ended free-text questions are scored by BLEU-4 (with smoothing, appropriate given the short length of typical VQA-RAD answers) and ROUGE-L F-measure. Both conditions are scored by the same metric implementation, so the two accuracy numbers in Section VI are computed identically.

D. Statistical Significance

Given VQA-RAD's test split is small (on the order of a few hundred examples), a raw accuracy delta between conditions could plausibly reflect sampling noise rather than a genuine effect. This is tested with a paired bootstrap: per-example scores from both conditions are resampled with replacement 10,000 times, preserving the pairing between a given test example's baseline and RAG scores, and a two-sided p-value together with a 95% confidence interval is reported for the mean delta. An effect is treated as reliable only when $p < 0.05$ and the confidence interval excludes zero.

E. Qualitative Error Analysis

Beyond the aggregate delta, every shared test example is categorized as *improved* (RAG correct, baseline incorrect), *regressed* (baseline correct, RAG incorrect), or unchanged (both correct or both incorrect), to characterize where retrieval helps or hurts rather than only whether it does so on average. Retrieved-evidence quality is profiled separately from answer correctness — mean top-1 retrieval similarity and the proportion of queries for which no evidence was retrieved at all — so that a weak result can be attributed to the retriever failing to find relevant images versus the model failing to make use of evidence it was given.

F. Hardware and Reproducibility

All experiments were run on a Kaggle Notebook with a single NVIDIA T4 GPU (15GB VRAM), using PyTorch 2.3.1 (with `transformers` upgraded to 4.46.3 and `tokenizers` to $\geq$0.20.3 to resolve a tokenizer deserialization incompatibility with the pinned `transformers` version) and faiss-cpu for index construction and search. The baseline vision-language model was loaded in half precision (float16) to fit comfortably within available GPU memory. Corpus embedding and index construction over the full ROCOv2 corpus (59,962 image-caption pairs after filtering) took approximately 17 minutes; each full evaluation run over the 451-example VQA-RAD official test split (baseline or a single retrieval-augmented condition) took approximately 2–3

TABLE II
BASELINE VS. RETRIEVAL-AUGMENTED RESULTS ON THE VQA-RAD TEST SPLIT.

Metric	Baseline	RAG	Delta
Closed-question accuracy	0.4303	0.4781	+0.0478
Open-question BLEU-4	0.0359	0.0389	+0.0030
Open-question ROUGE-L	0.1422	0.1428	+0.0006
Overall exact match	0.2683	0.2949	+0.0266

TABLE III
PAIRED BOOTSTRAP SIGNIFICANCE TEST RESULTS.

Comparison	Delta	95% CI	p	Sig. ($p < 0.05$)
closed_exact_match	+0.0478	[-0.0199, 0.1155]	0.1848	No
open_rouge_l	+0.0006	[-0.0321, 0.0328]	0.9642	No

minutes. Every number reported in Section VI is generated by the released code (`scripts/run_baseline.py`, `scripts/run_rag_eval.py`, `scripts/run_experiments.py`, `scripts/generate_paper_results.py`) against a fixed, versioned configuration (`configs/model_config.yaml`), so any of these figures can be regenerated by re-running the corresponding script.

VI. RESULTS

Table II reports headline metrics for both conditions on the VQA-RAD test split, computed by the identical scoring code (`src/evaluation/metrics.py`) for both.

Table III reports the paired bootstrap significance test (10,000 resamples, 95% confidence interval) for each comparison, computed by `src/evaluation/significance.py`.

Table IV reports the top-k sweep from `scripts/run_experiments.py`.

Table II shows the retrieval-augmented condition outperformed the baseline on every reported metric at $k = 5$: closed-question accuracy rose from 43.0% to 47.8% (+4.8 points), overall exact match from 26.8% to 29.5% (+2.7 points), with smaller gains on open-question BLEU-4 and ROUGE-L. Table III reports the paired bootstrap significance test on both the closed-question exact-match delta and the open-question ROUGE-L delta; neither reaches significance at $\alpha = 0.05$ ($p = 0.1848$ and $p = 0.9642$ respectively, both confidence intervals spanning zero). The closed-question result is the more suggestive of the two – its p-value is closer to conventional significance thresholds and its confidence interval's lower bound is close to zero – but with $n = 251$ closed-question examples we cannot reject the null hypothesis that this delta arose by chance. The open-question ROUGE-L delta is close to zero in absolute terms and correspondingly far from significant. Table IV shows the same pattern observed at $k = 5$ recurs across the sweep: closed accuracy is highest at $k = 1$ (54.98%) and declines monotonically through $k = 10$ (46.61%), a pattern discussed further in Section VII.

TABLE IV
RETRIEVAL-AUGMENTED ACCURACY AS A FUNCTION OF TOP-k.

k	Closed Acc.	Open BLEU	Open ROUGE-L
1	0.5498	0.0412	0.1492
3	0.4900	0.0392	0.1447
5	0.4781	0.0389	0.1428
10	0.4661	0.0399	0.1516

VII. DISCUSSION

This section addresses three questions directly from the results in Section VI and the error-analysis output in `docs/experiments_findings.md`.

A. Does retrieval help overall, and is the effect reliable?

At the default retrieval depth ($k = 5$), the retrieval-augmented condition outperformed the baseline on every metric: closed-question accuracy rose from 43.0% to 47.8% (+4.8 points), overall exact match from 26.8% to 29.5% (+2.7 points), with smaller gains on BLEU-4 and ROUGE-L. However, the paired bootstrap significance test on the closed-question delta returned $p = 0.1848$ with a 95% confidence interval of $[-0.020, 0.116]$, and the open-question ROUGE-L delta returned $p = 0.9642$ with a confidence interval of $[-0.032, 0.033]$ – both spanning zero. With $n = 451$ test examples (251 closed, 200 open), we cannot reject the null hypothesis that either improvement arose by chance, though the closed-question result is the more suggestive of the two. The honest reading is that retrieval augmentation produced a consistent, direction-positive effect across all reported metrics, but the test set is too small, and the effect too modest, to confirm it is real rather than sampling noise.

B. Why does performance peak at $k = 1$ rather than at higher retrieval depths?

The top-k sweep (Table IV, Section VI) shows a monotonic decline in closed-question accuracy as k increases from 1 to 10: accuracy falls from 55.0% at $k = 1$ to 46.6% at $k = 10$, while open-question BLEU-4 and ROUGE-L do not show as clean a monotonic trend (ROUGE-L is in fact highest at $k = 10$, 0.1516). The clearest and most interpretable effect in the sweep is on closed-question accuracy. A plausible explanation is evidence dilution: mean top-1 retrieval similarity was high (0.971), indicating the single most similar corpus image is usually a strong match, but each additional retrieved caption beyond the first introduces topically related but increasingly less relevant text into the prompt. For a base (not instruction-tuned) checkpoint such as `blip2-opt-2.7b`, a longer, noisier evidence block may compete with the actual question for the model's limited effective context rather than reinforcing it. An equally plausible alternative explanation, not distinguishable from evidence dilution using this experiment alone, is that the retriever itself is question-agnostic: because it embeds only the query image (Section IV) and never the question text, only the single most visually similar corpus entry is reliably relevant to what is actually being asked, and captions beyond the first are increasingly likely to be irrelevant to the specific question rather than merely diluted – a distinction future work with question-aware retrieval (Section VIII) could disentangle.

C. What do the improved/regressed error categories show?

Of the 451 test examples, retrieval changed the outcome for a similar number of cases in each direction: 54 examples improved (baseline wrong, RAG correct) versus 44 that regressed (baseline correct, RAG wrong), a near-even split that is consistent with the modest, non-significant aggregate delta reported above. The dominant category, however, was *both wrong* (260 of 451 examples, 58%), with only 93 examples (21%) answered correctly by both conditions. This indicates that the baseline VLM's zero-shot capability on radiology imagery, not the presence or absence of retrieved evidence, is the primary bottleneck on this benchmark: retrieval can only change the outcome on the minority of examples where the underlying model was close to a correct answer in the first place.

VIII. Limitations

A. Note on a Corrected Evaluation Pipeline

An earlier version of this evaluation pipeline contained two bugs, corrected prior to the results reported in Section VI. First, the closed/open answer-type label was read from a field that does not exist on the Hugging Face mirror of VQA-RAD used here, causing every example to be silently mislabeled open-ended and closed-question accuracy to be permanently reported as 0.0000. Second, the evaluation set was constructed by internally re-splitting VQA-RAD's official training portion (yielding a 180-example test set) rather than using VQA-RAD's own official 451-example held-out test split, which is what other published VQA-RAD baselines evaluate against. Both are fixed in the pipeline this paper's numbers are drawn from: answer type is recovered from the answer text (Section III), and the official test split is used directly. We document this correction explicitly, rather than silently revising the numbers, in the interest of transparent research practice.

B. Dataset Scope

Both datasets used here are deliberately non-credentialed (Section III), which was necessary for a timeline-constrained project but has real consequences for generalizability. VQA-RAD's test split is small (on the order of a few hundred examples), which is why Section V treats statistical significance as a requirement rather than an optional check, but a small test split also limits the granularity of any subgroup analysis (e.g. by modality or anatomical region) that a larger benchmark would support. ROCOv2's captions are figure captions drawn from published literature, not full clinical reports, and are English-only; neither dataset was designed as a matched pair, so the retrieval corpus and evaluation benchmark may differ somewhat in imaging style, patient population, or reporting convention. MIMIC-CXR, the credentialed dataset most published medical VLM work uses, is a natural extension once credentialed access is obtained (Section IX).

C. Retriever Design

The retriever embeds and matches only the query *image*: the CLIP image encoder retrieves corpus entries by visual similarity alone, and the VQA question text never participates in retrieval (Section IV). As discussed in Section VII, this is a plausible alternative explanation (alongside evidence dilution) for why accuracy declines as k increases – a question-aware or cross-modal retriever, which this work does not implement, could retrieve evidence relevant to what is specifically being asked rather than only to what the image visually resembles. The retriever also uses a single, off-the-shelf CLIP checkpoint (`openai/clip-vit-base-patch32`) trained on general-domain web images, not fine-tuned on medical imagery specifically. Whether a medical-domain-adapted image encoder would retrieve more clinically relevant evidence, and thereby change the retrieval-augmented result, is untested here. Likewise, only an exact flat inner-product index is used; this is an appropriate, deliberate choice at ROCOv2's corpus scale ($\sim$90K vectors, Section IV) but would not scale without revisiting approximate indexing (IVF/HNSW) at a larger corpus size, and the top-k sweep in Section VI covers only a handful of discrete values, not a continuous sensitivity analysis.

D. Evaluation Methodology

Closed-question accuracy in this paper relies on an answer-type label recovered from the answer text (Section III: a normalized answer of exactly "yes" or "no" is treated as closed-ended), since the VQA-RAD mirror used here does not expose the original annotation. This recovers the large majority of the true closed/open split but will mislabel a small number of closed-ended, non-yes/no questions as open-ended; the reported closed-question accuracy should be read with this approximation in mind. Open-ended question scoring relies on BLEU-4 and ROUGE-L, both surface n-gram overlap metrics that can penalize a correct answer phrased differently from the reference and, conversely, reward superficial lexical overlap without semantic correctness. BLEU-4 in particular is close to structurally uninformative at VQA-RAD's typical open-ended answer length (often 3–4 tokens): even two identical strings can score well below 1.0 under BLEU-4 with smoothing, since a short answer offers few or no 4-gram matches regardless of correctness; the BLEU-4 deltas reported in Section VI should be weighted accordingly relative to ROUGE-L and exact match. The 0.5 ROUGE-L threshold used to bucket open-ended examples into *improved/regressed* categories for qualitative review (Section V) is a convenience heuristic for categorization, not a validated correctness threshold, and is not used anywhere in the headline reported metrics. No clinician review of model answers for clinical adequacy (as opposed to lexical match against a reference) was performed; this is a meaningful gap between what this evaluation measures and what would be required before any claim of clinical utility.

E. Compute Constraints

As a resource-constrained, timeline-bound project, this work evaluates a single baseline checkpoint family (`Salesforce/blip2-opt-2.7b`, with `microsoft/llava-med-v1.5-mistral-7b` as an alternate candidate) at a single scale, rather than an ablation across multiple model sizes or architectures. Similarly, only one prompt template (`configs/model_config.yaml`'s `rag.evidence_instruction`) was used for the retrieval-augmented condition; prompt phrasing is known in the broader RAG literature to affect how well a model makes use of retrieved context, and this was not itself an experimental variable here.

F. Scope of Claims

This is a research prototype intended to answer a narrow empirical question — does retrieval augmentation change VLM accuracy on a specific medical VQA benchmark, under one model and retriever configuration — and is explicitly not a validated diagnostic or clinical decision-support tool. Any result reported in Section VI should be read as evidence about this specific pipeline and benchmark, not as a general claim about retrieval-augmented medical VLMs as a class.

IX. Conclusion and Future Work

This paper presented a controlled comparison between a vision-language model answering medical visual questions from its parameters alone and the same model prompted with evidence retrieved from a CLIP-embedded FAISS index over the ROCOv2 corpus, built entirely on openly licensed, non-credentialed data and evaluated on VQA-RAD's official held-out test split with paired statistical significance testing. Retrieval augmentation produced a consistent, direction-positive improvement over the non-augmented baseline on all reported metrics at $k = 5$ (closed-question accuracy +4.8 points, overall exact match +2.7 points), but a paired bootstrap test found neither improvement statistically significant at $\alpha = 0.05$ (closed-question exact match: $p = 0.1848$, 95% CI $[-0.020, 0.116]$; open-question ROUGE-L: $p = 0.9642$, 95% CI $[-0.032, 0.033]$; $n = 451$). A secondary finding from the top-k sweep — that closed-question accuracy is highest at $k = 1$ and degrades monotonically as more evidence is retrieved — suggests that, for this baseline checkpoint, retrieval depth introduces a trade-off between evidence coverage and prompt noise that is at least as important as the presence of retrieval itself.

A. Future Work

Four extensions follow directly from the limitations in Section VIII. First, extending the retrieval corpus to include MIMIC-CXR once credentialed access is obtained would test whether this paper's finding holds on a corpus of matched clinical reports rather than figure captions, and at substantially larger scale. Second, a medical-domain-adapted image encoder in place of general-domain CLIP could be evaluated as a retriever, to isolate whether retrieval quality itself was a limiting factor. Third, a question-aware or cross-modal retriever – one that embeds the question text alongside the query image, rather than the image alone – could disentangle whether the top-k sweep's declining accuracy reflects evidence dilution, retrieval irrelevance, or both. Fourth, a small clinician review of a sample of model answers — assessing clinical adequacy rather than lexical overlap with a reference answer — would address the gap between this paper's automatic metrics and any claim of clinical utility, and is a necessary step before this pipeline could inform any real decision-support use case.

The complete, tested, config-driven codebase for this pipeline — dataset acquisition, baseline and retrieval-augmented models, significance testing and error analysis, and a FastAPI deployment — is released alongside this paper to support exactly these extensions.

References

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